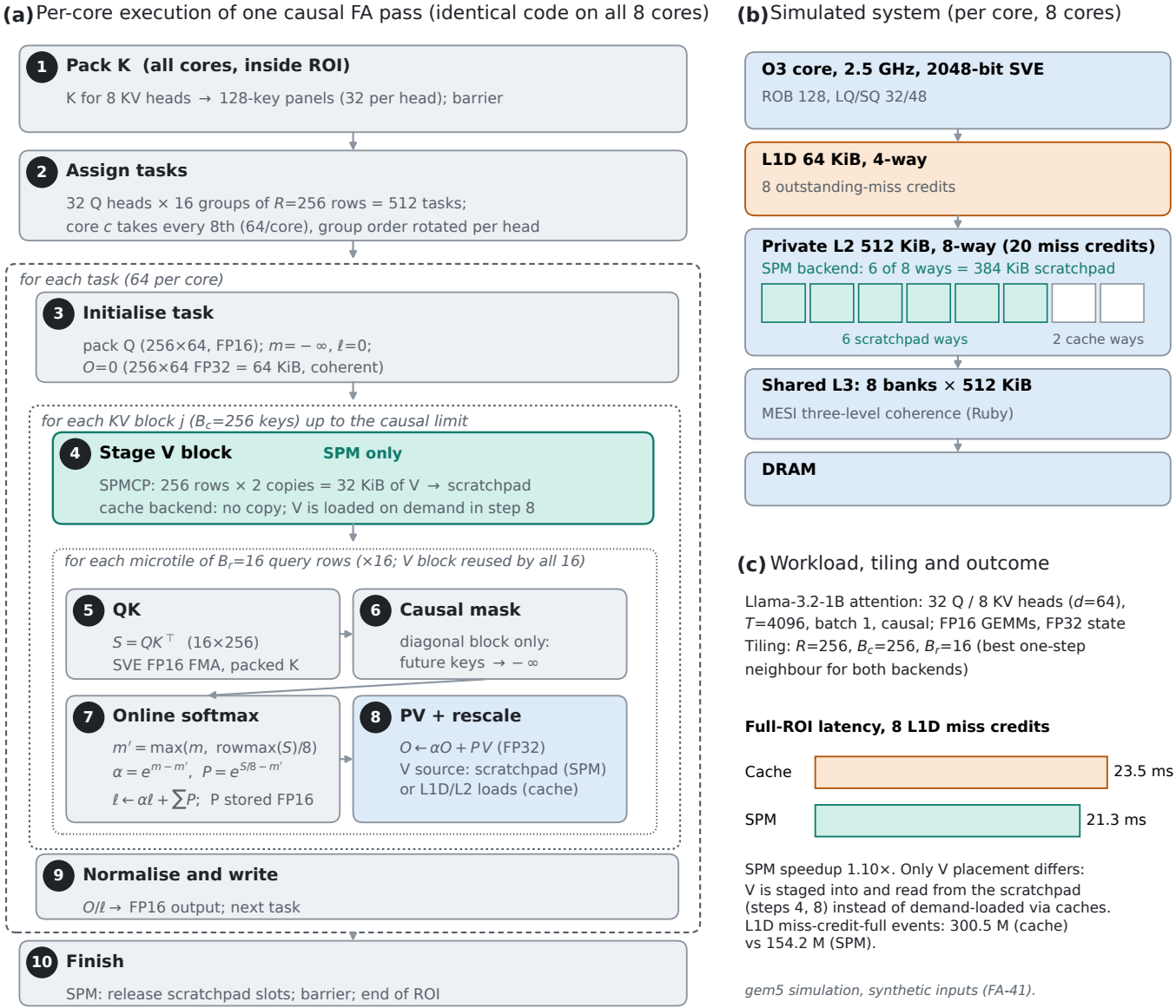

(a) Per-core execution of one causal FA pass (identical code on all 8 cores)
1 Pack K (all cores, inside ROI)
K for 8 KV heads → 128-key panels (32 per head); barrier
2 Assign tasks
32 Q heads × 16 groups of R=256 rows = 512 tasks;
core c takes every 8th (64/core), group order rotated per head
for each task (64 per core)
3 Initialise task
pack Q (256×64, FP16); m=−∞, ℓ=0;
O=0 (256×64 FP32 = 64 KiB, coherent)
for each KV block j (B_c=256 keys) up to the causal limit
4 Stage V block
SPM only
SPMCP: 256 rows × 2 copies = 32 KiB of V → scratchpad
cache backend: no copy; V is loaded on demand in step 8
for each microtile of B_r=16 query rows (×16; V block reused by all 16)
5 QK
S = QK^⊤ (16×256)
SVE FP16 FMA, packed K
6 Causal mask
diagonal block only:
future keys → −∞
7 Online softmax
m′ = max(m, rowmax(S)/8)
α = e^{m−m′}, P = e^{S/8−m′}
ℓ ← αℓ + ∑P; P stored FP16
8 PV + rescale
O ← αO + PV (FP32)
V source: scratchpad (SPM)
or L1D/L2 loads (cache)
9 Normalise and write
O/ℓ → FP16 output; next task
10 Finish
SPM: release scratchpad slots; barrier; end of ROI
(b) Simulated system (per core, 8 cores)
O3 core, 2.5 GHz, 2048-bit SVE
ROB 128, LQ/SQ 32/48
L1D 64 KiB, 4-way
8 outstanding-miss credits
Private L2 512 KiB, 8-way (20 miss credits)
SPM backend: 6 of 8 ways = 384 KiB scratchpad
6 scratchpad ways
2 cache ways
Shared L3: 8 banks × 512 KiB
MESI three-level coherence (Ruby)
DRAM
(c) Workload, tiling and outcome
Llama-3.2-1B attention: 32 Q / 8 KV heads (d=64),
T=4096, batch 1, causal; FP16 GEMMs, FP32 state
Tiling: R=256, B_c=256, B_r=16 (best one-step
neighbour for both backends)
Full-ROI latency, 8 L1D miss credits
Cache
23.5 ms
SPM
21.3 ms
SPM speedup 1.10×. Only V placement differs:
V is staged into and read from the scratchpad
(steps 4, 8) instead of demand-loaded via caches.
L1D miss-credit-full events: 300.5 M (cache)
vs 154.2 M (SPM).
gem5 simulation, synthetic inputs (FA-41).